

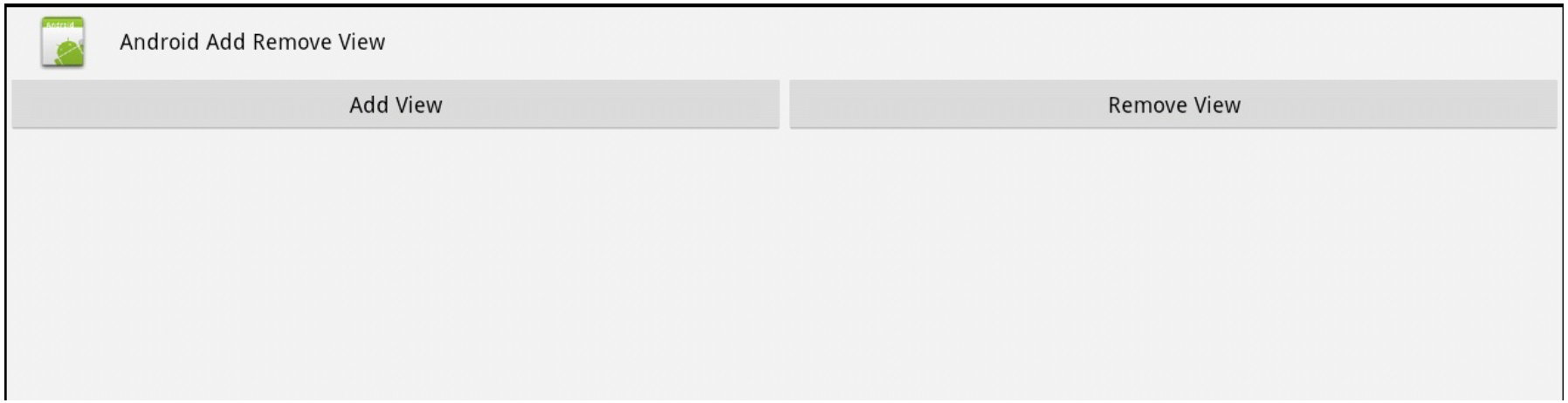
List-Based Widgets (Continued)

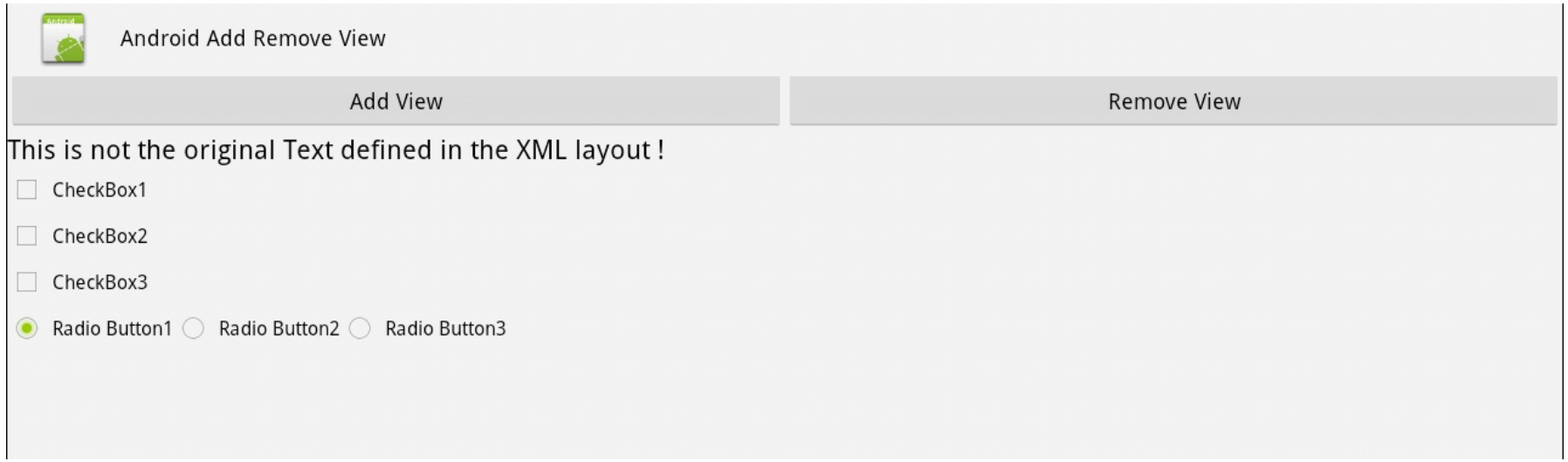
ListView Widget

- **LayoutInflater**
- Advance custom-made ListView
  - ArrayAdapter
  - BaseAdapter
- Spinner
- ExpandableListView

LayoutInflater

- The LayoutInflater class converts an XML layout specification into an actual tree of View objects.





LayoutInflater

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
        ....
        android:id="@+id/root_view">
</LinearLayout>
```

LayoutInflater

```
ViewGroup root =
        (ViewGroup)findViewById(R.id.root_view);
LayoutInflater inflater = getLayoutInflater();
inflater.inflate(R.layout.child_view, root);
```

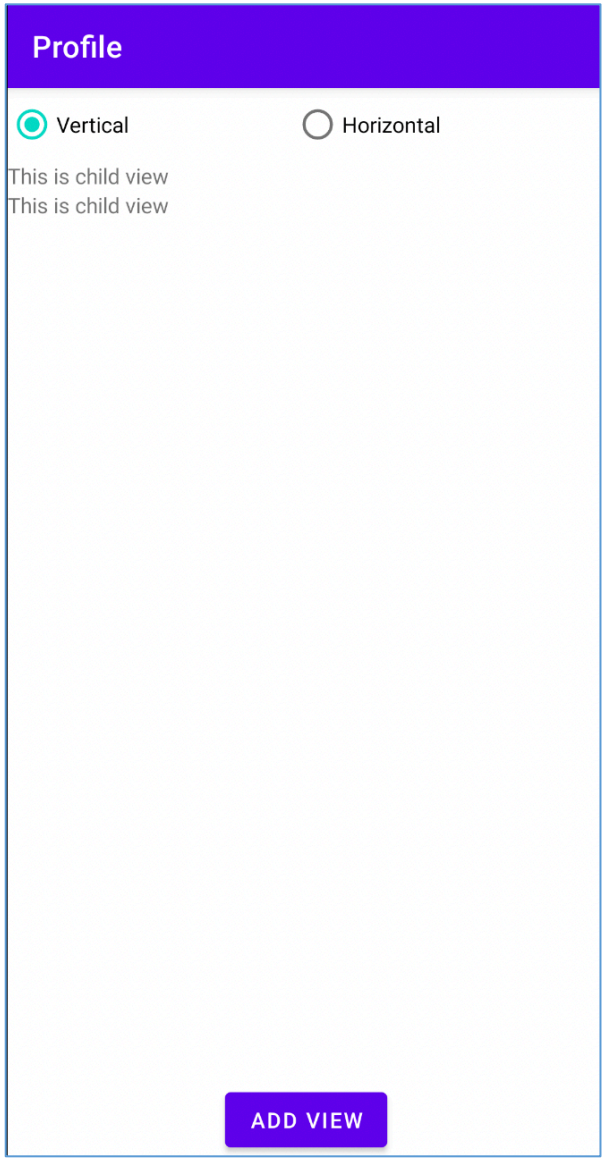
LayoutInflater

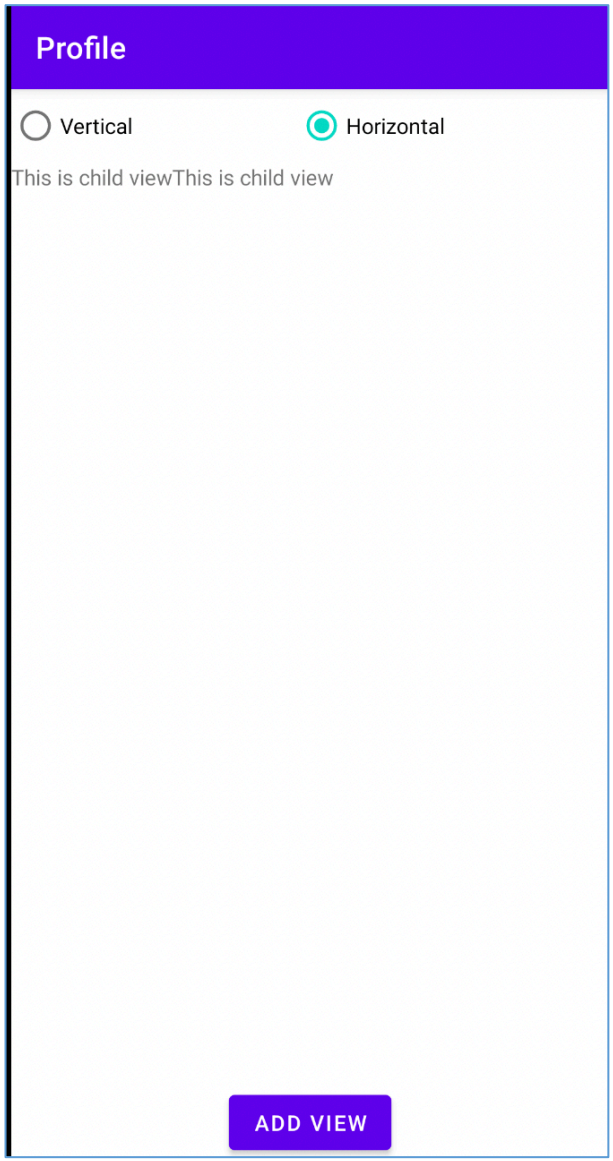
```
ViewGroup root =
    (ViewGroup)findViewById(R.id.root_view);

LayoutInflater inflater = getLayoutInflater();
View childView =
    inflater.inflate(R.layout.child_view, null);

root.addView(childView);
```

LayoutInflater





ListView Widget

- LayoutInflater
- **Advance custom-made ListView**
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Custom-made List-Based Widgets (Create your own Data Adapter class)

In order to customize a Data Adapter, you need to:

1. Create a class extending the concrete **ArrayAdapter** class
2. Override its **getView**(), and
3. Construct (inflate) your rows yourself.

```
public class MyCustomAdapter extends ArrayAdapter{

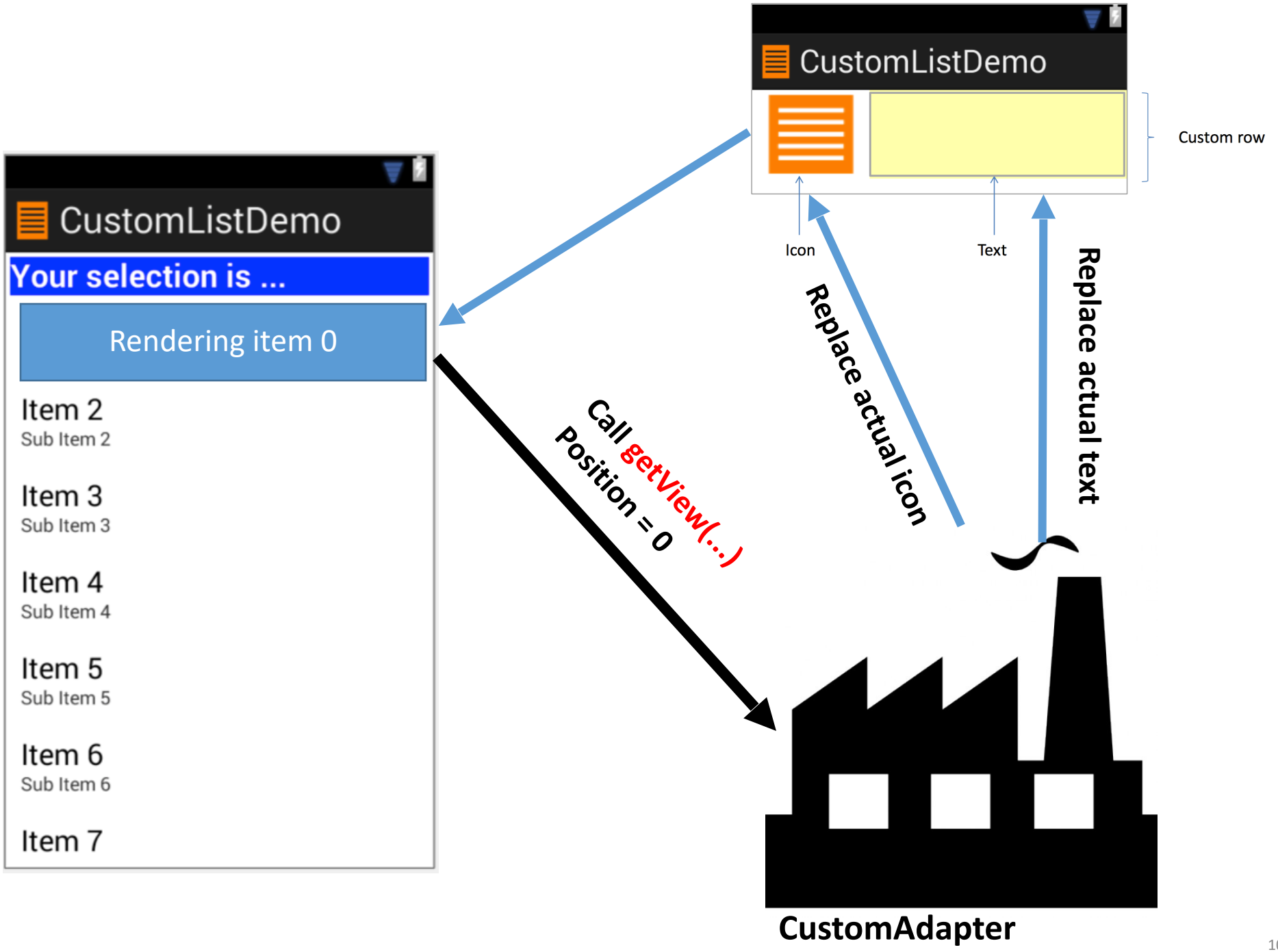
   // class variables go here ...

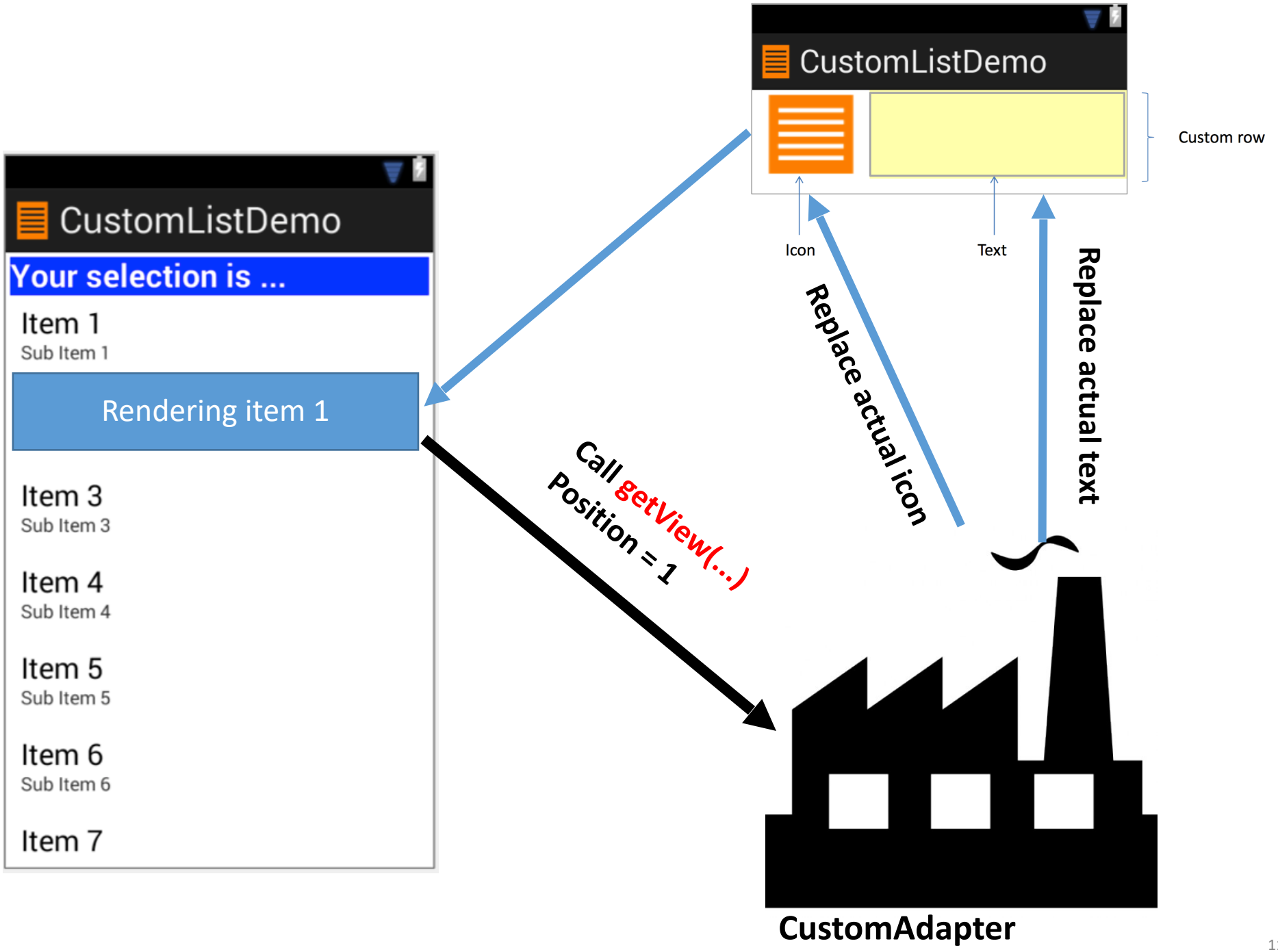
   public MyCustomAdapter(...) {  }

   public View getView(...) {    }

}//MyCustomAdapter
```

For each data element supplied by the adapter, the `getView()` method returns its corresponding 'visible' View.


CustomListDemo
Your selection is ...
Rendering item 0
Item 2
Sub Item 2
Item 3
Sub Item 3
Item 4
Sub Item 4
Item 5
Sub Item 5
Item 6
Sub Item 6
Item 7
CustomListDemo
Custom row
Icon
Text
Replace actual icon
Replace actual text
Call getView(...)
Position = 0
CustomAdapter


CustomListDemo
Your selection is ...
Item 1
Sub Item 1
Rendering item 1
Item 3
Sub Item 3
Item 4
Sub Item 4
Item 5
Sub Item 5
Item 6
Sub Item 6
Item 7
CustomListDemo
Custom row
Icon
Text
Replace actual icon
Replace actual text
Call getView(...)
Position = 1
CustomAdapter

Custom-made List-Based Widgets (Create your own Data Adapter class)

```
private class MyCustomAdapter extends
                                ArrayAdapter<String> {
    public MyCustomAdapter(...) { ... }

    @override
    public View getView(...) { ... }
}

adapter = new MyCustomAdapter(
                    this,
                    R.layout.my_list_item,
                    data);

listView.setAdapter(adapter);
```

Custom-made List-Based Widgets (Inline override getView method)

```
adapter = new ArrayAdapter<String>(
                    this,
                    R.layout.my_list_item,
                    data) {

    @override
    public View getView(...) { ... };

};

listView.setAdapter(adapter);
```

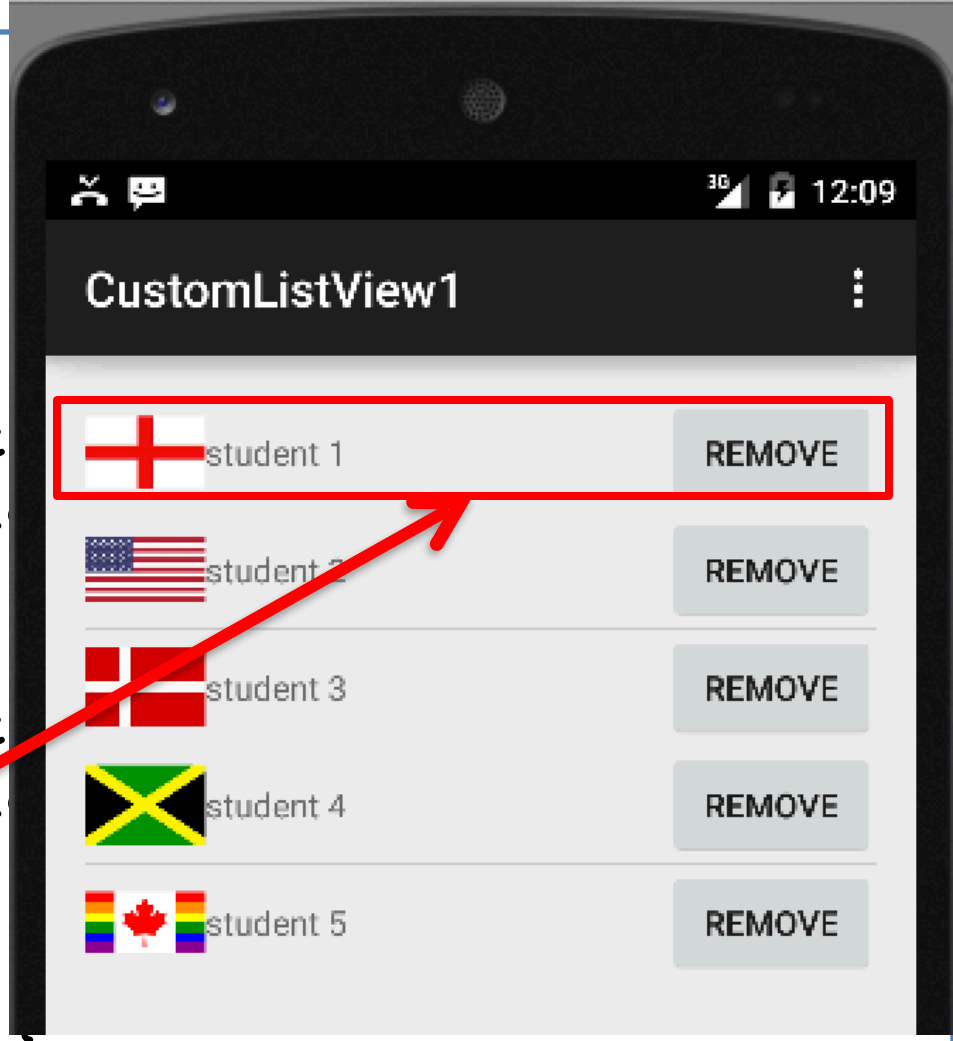
ListView Widget

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- Spinner
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Custom-made List-Based Widgets (BaseAdapter class)

```
public class MyAdapter extends BaseAdapter {
    public int getCount() {    }
    public Object getItem(...) {    }
    public long getItemId(...) {    }
    public View getView(...) {    }
}
```

```
@Override
public int getCount() {
    return data.size();
}
@Override
public Object getItem(int
    return data.get(positi
}
@Override
public long getItemId(int
    return data.get(positi
}
@Override
public View getView(...)
    ...
}
```



BaseAdapter: getView(…)

```
@Override
public View getView(int position,
                    View convertView,
                    ViewGroup parent) {
    TextView textView;

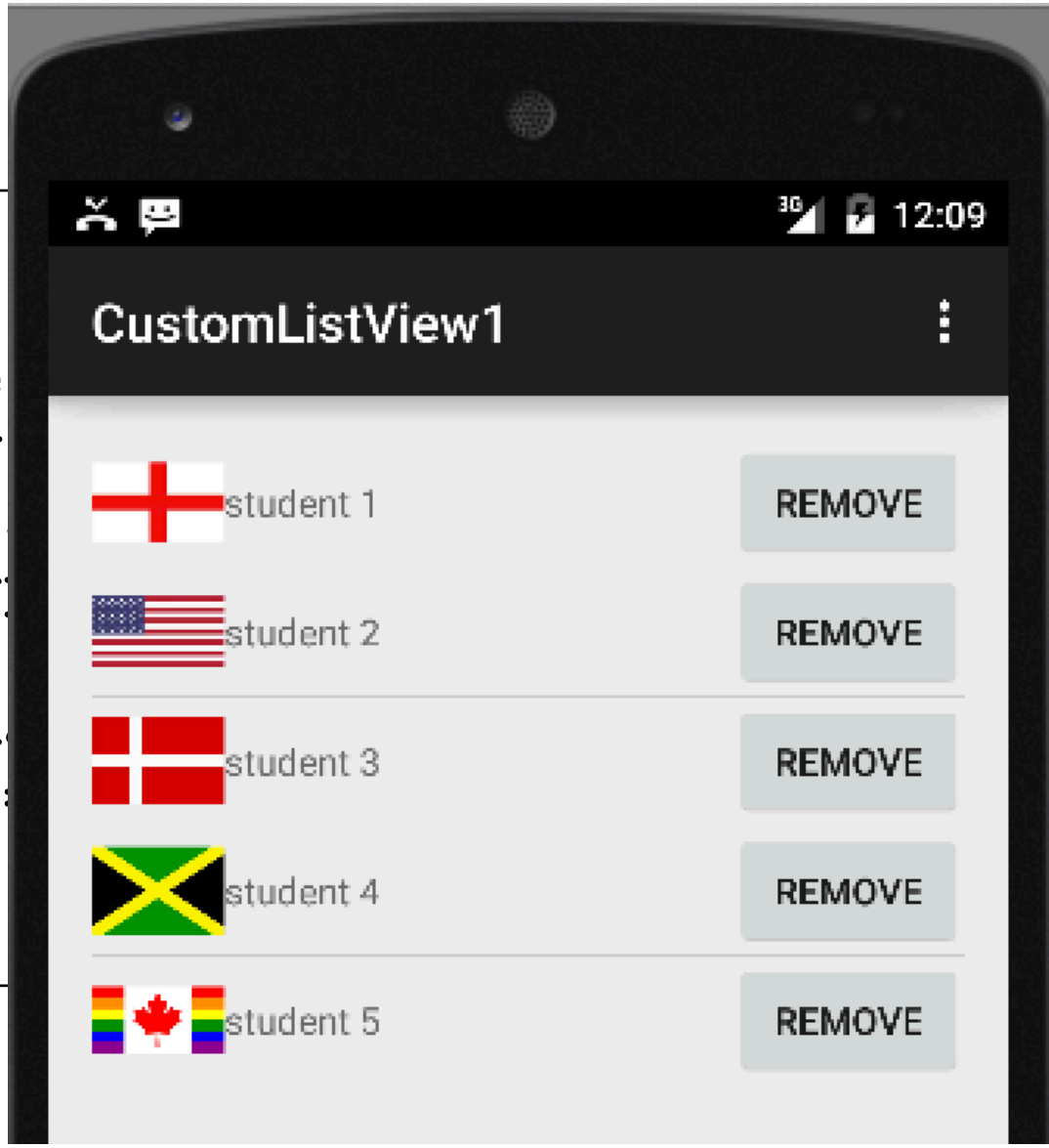
    LayoutInflater inflater =
                    LayoutInflater.from(context);
    View itemView = inflater.inflate(
                        R.layout.my_list_item, null);

    //do something with itemView

    return itemView;
}
```

Example 01

```
public class Student
   private int id;
   private String name
   private int country
   public Student(int
                  countr
          this.id = id;
          this.name = n
          this.country
  }
}
```



Example 02



Html.fromHtml("°") - http://www.ascii.cl/htmlcodes.htm

ListView Widget

- LayoutInflater
- Advance custom-made ListView
  - ArrayAdapter
  - BaseAdapter
- **Spinner**
  - **Simple Spinner**
  - Custom Spinner 1
  - Custom Spinner 2
- ExpandableListView

Spinner
(Simple Spinner)

```
public class SimpleSpinner extends Activity {
    private Spinner spinner;
    private ArrayList<String> dataProvider;
    private ArrayAdapter adapter;
    ...
```

Spinner
(Simple Spinner)

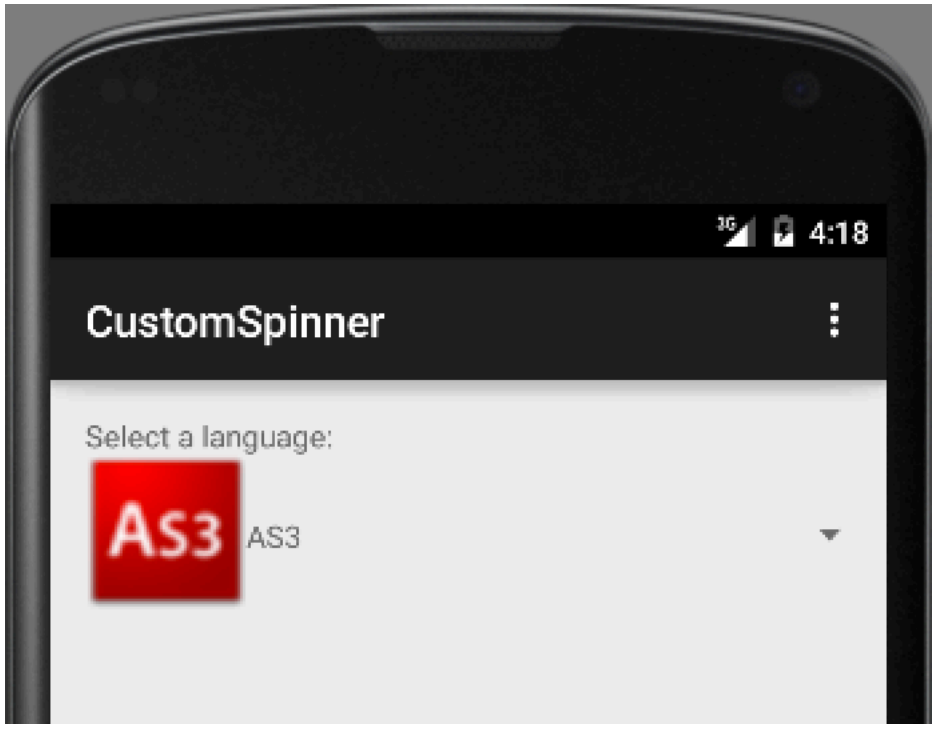
```
@Override
protected void onCreate(Bundle savedInstanceState) {
  super.onCreate(savedInstanceState);
  setContentView(R.layout.activity_simple_spinner);

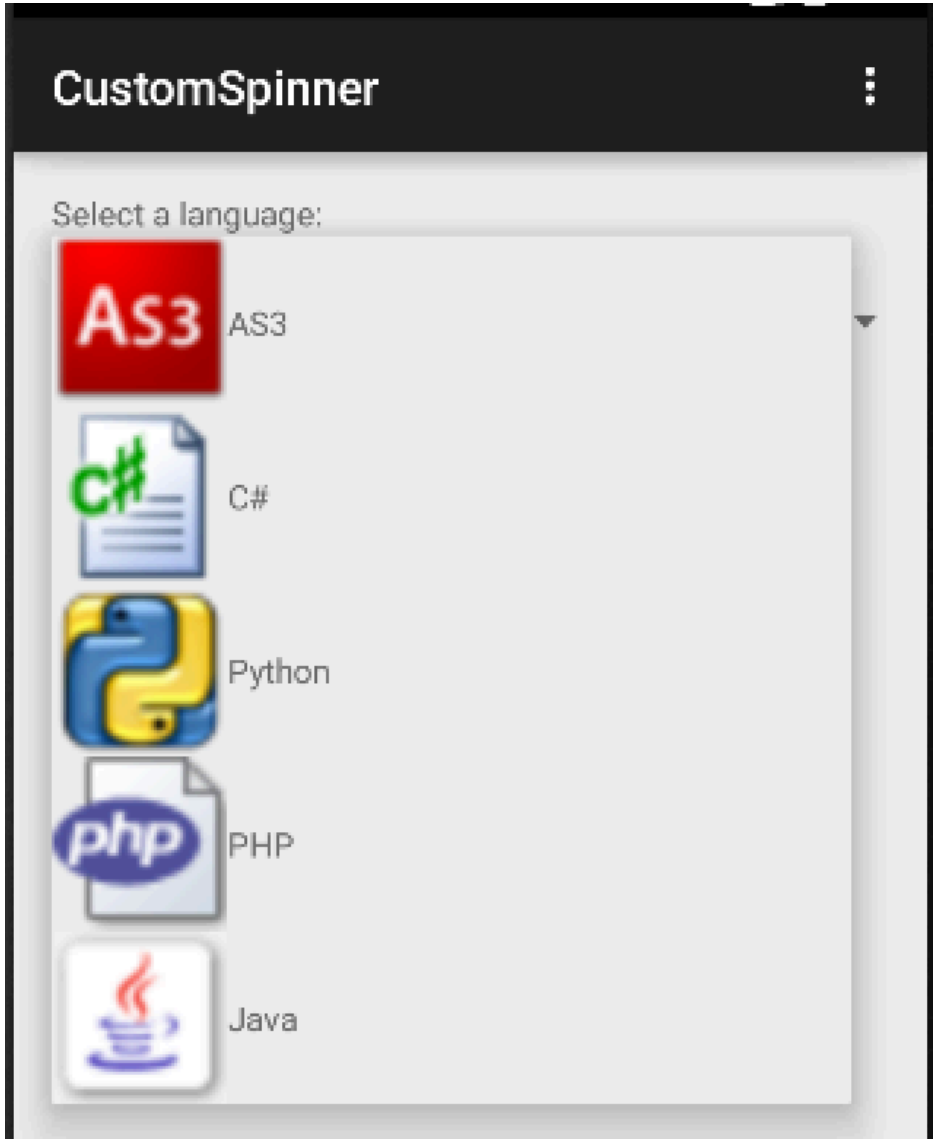
  dataProvider = new ArrayList<>(Arrays.asList(
         "Ha Noi", "Ho Chi Minh", "Binh Duong",
         "Da Nang", "Quang Ngai", "Hai Phong",
         "Binh Phuoc"));
  spinner = (Spinner) findViewById(R.id.spinner);
  adapter = new ArrayAdapter(this,
         android.R.layout.simple_spinner_item,
         dataProvider);
  spinner.setAdapter(adapter);
}
```

ListView Widget

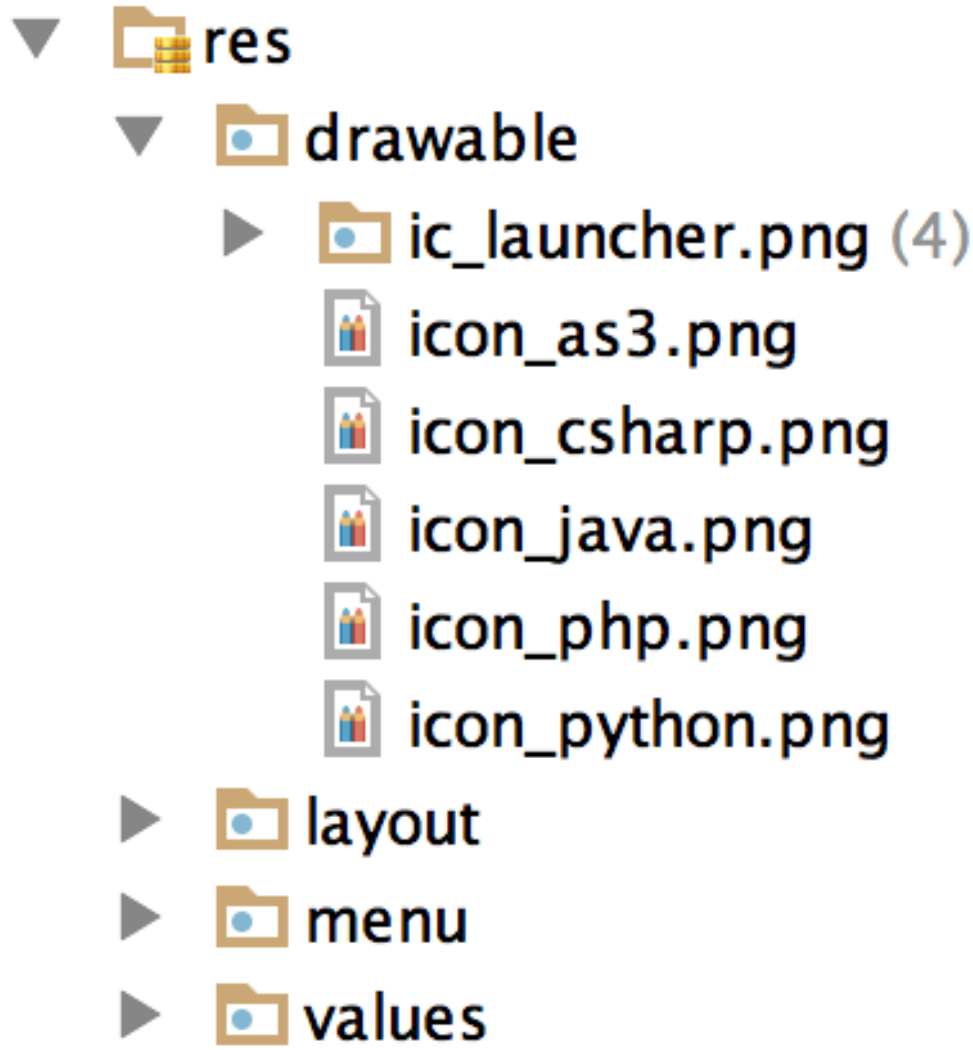
- LayoutInflater
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Custom Spinner 1





Custom Spinner 1



Custom Spinner 1

```
public class LanguageInfo {
    private String name;
    private int icon;
    public LanguageInfo(String name, int icon) {
        this.name = name;
        this.icon = icon;
    }
    ... Getter/Setter here ...
    @Override
    public String toString() {
        return name;
    }
}
```

Custom Spinner 1 (CustomSpinner.java)

```
public class CustomSpinner
             extends ActionBarActivity {
    private Spinner spinner;
    private ArrayList<LanguageInfo> dataProvider;
    private CustomAdapter adapter;
    ...
```

Custom Spinner 1 (onCreate – CustomSpinner.java)

```
dataProvider = new ArrayList<>(Arrays.asList(
     new LanguageInfo("AS3",
                    R.drawable.icon_as3),
     new LanguageInfo("C#",
                    R.drawable.icon_csharp),
     new LanguageInfo("Python",
                    R.drawable.icon_python),
       new LanguageInfo("PHP",
                R.drawable.icon_php),
     new LanguageInfo("Java",
                    R.drawable.icon_java)
       ));
spinner = (Spinner) findViewById(R.id.spinner);
adapter = new CustomAdapter(dataProvider, this);
spinner.setAdapter(adapter);
```

Custom Spinner 1 (CustomAdapter.java)

```
@Override
public View getView(int position,
                View convertView, ViewGroup parent) {
   LayoutInflater inflater =
                    LayoutInflater.from(context);
   View row = inflater.inflate(
                R.layout.custom_spinner_item_1, null);
   LanguageInfo info = dataProvider.get(position);
   ImageView imageView = (ImageView)
         row.findViewById(R.id.imagePlaceHolder);
   TextView textView = (TextView)
         row.findViewById(R.id.textPlaceHolder);
   imageView.setImageResource(info.getIcon());
   textView.setText(info.getName());
   return row;
}
```

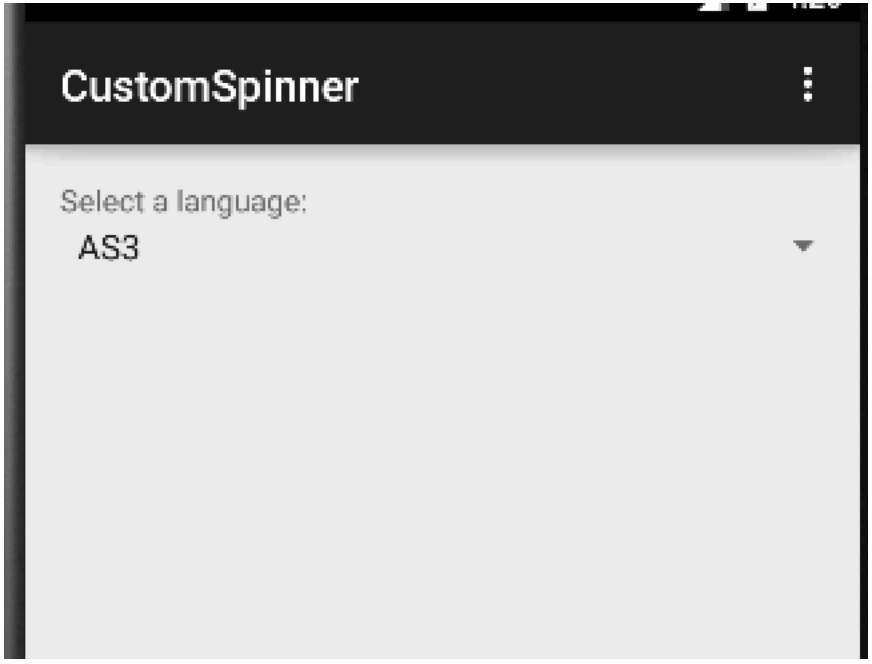
Custom Spinner 1 (layout_main.xml)

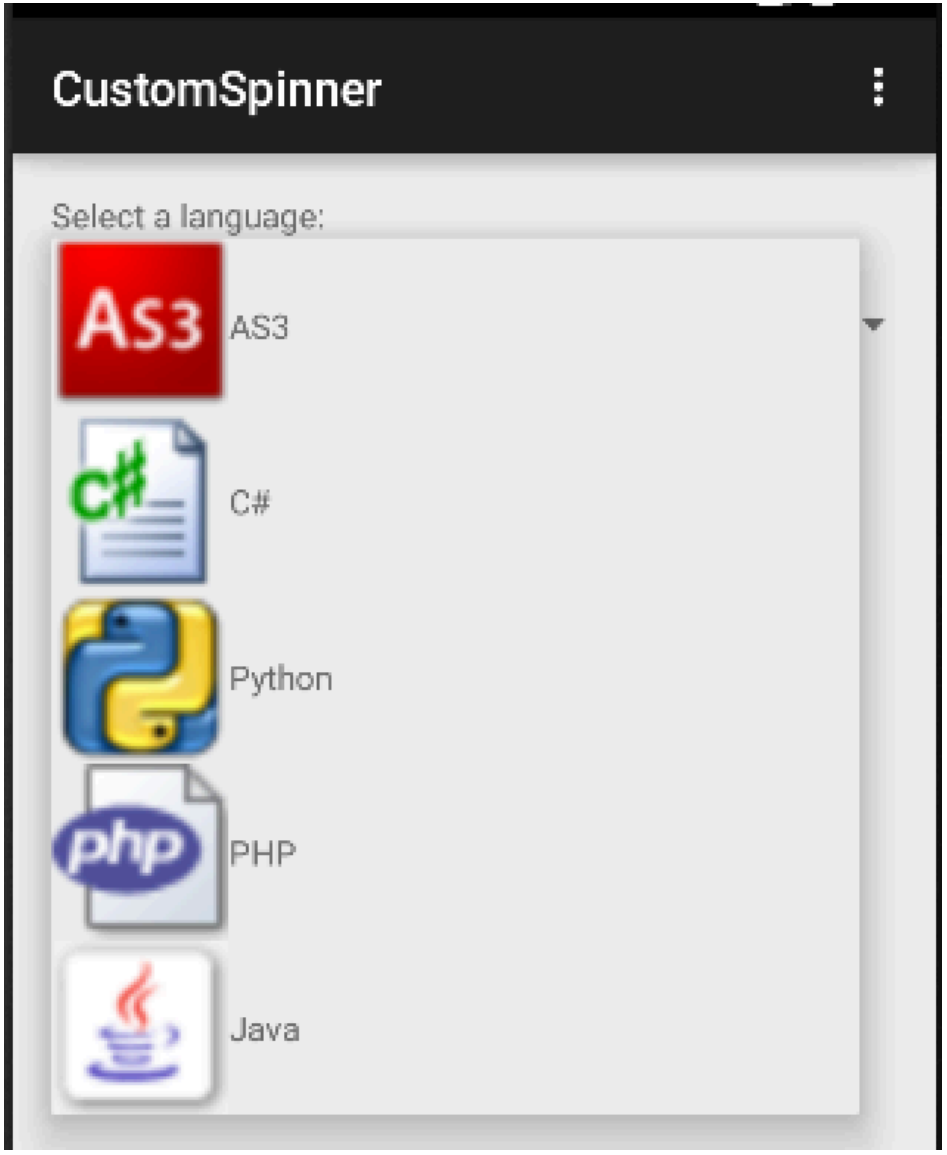
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout>
   <ImageView
      android:layout_width="74dp"
      android:layout_height="73dp"
      android:id="@+id/imagePlaceHolder"
      android:layout_gravity="center_vertical"/>
   <TextView
      android:layout_width="0dp"
      android:layout_height="wrap_content"
      android:layout_weight="1"
      android:text="Text will be replaced here"
      android:id="@+id/textPlaceHolder"
      android:layout_gravity="center_vertical" />
</LinearLayout>
```

ListView Widget

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  - **Custom Spinner 2**
- ExpandableListView

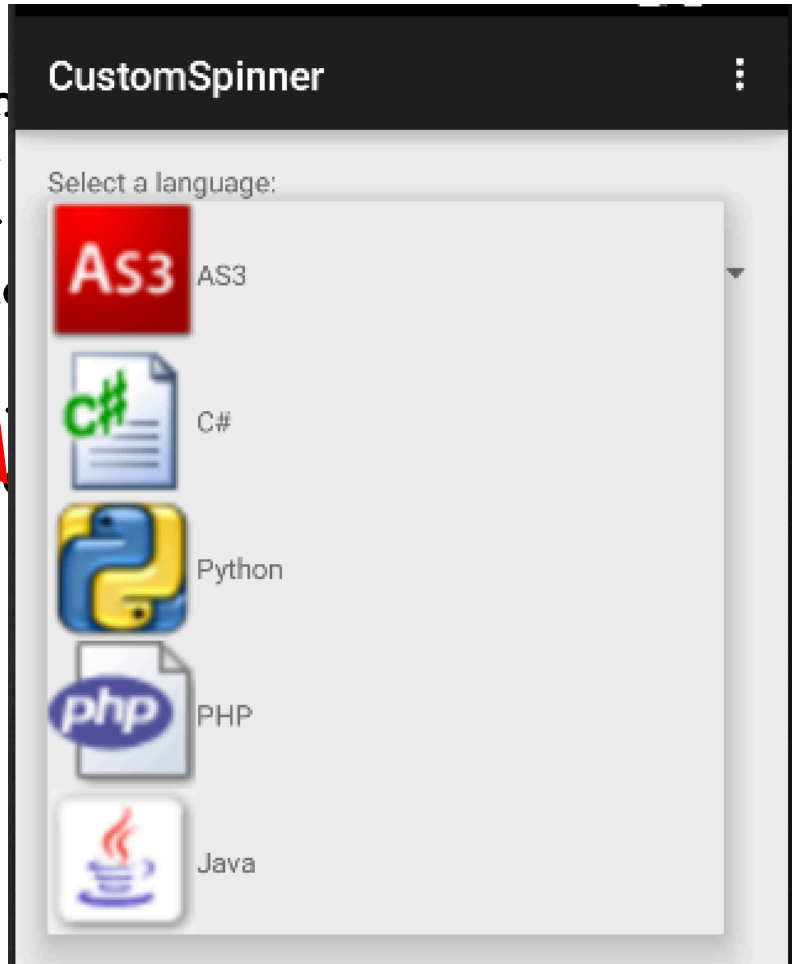
Custom Spinner 2





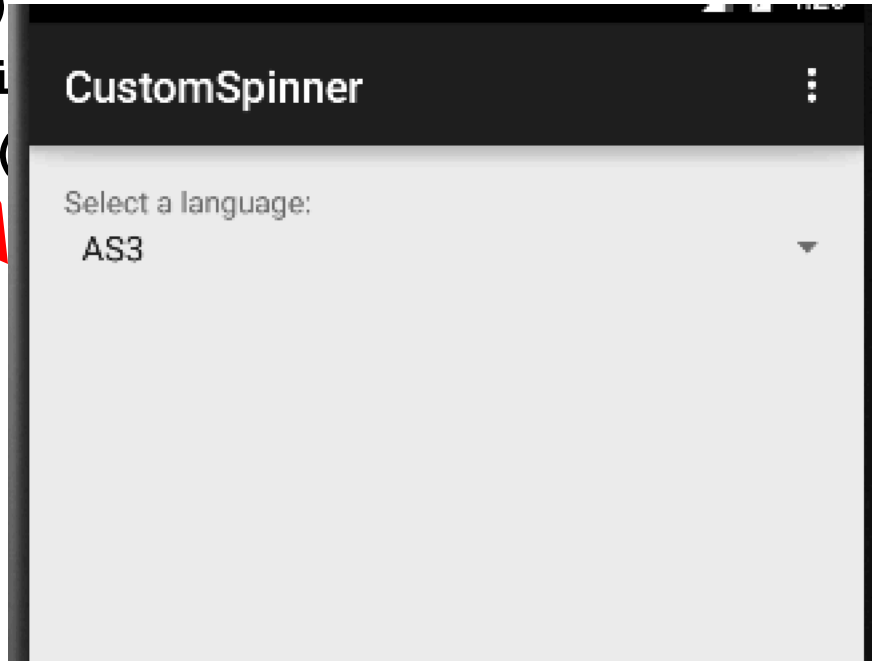
Custom Spinner 2 (CustomAdapter.java)

```
@Override
public View getDropDownView(int position,
                View convertView, ViewGroup parent) {
  LayoutInflater inflater = LayoutInflater.from(context);
  View row = inflater.inflate(
                R.layout.custom_spinn
  LanguageInfo info = dataProvider
  ImageView imageView = (ImageView
         row.findViewById(R.id.ima
  TextView textView = (TextView)
         row.findViewById(R.id.tex
  imageView.setImageResource(info.
  textView.setText(info.getName())
  return row;
}
```



Custom Spinner 2 (CustomAdapter.java)

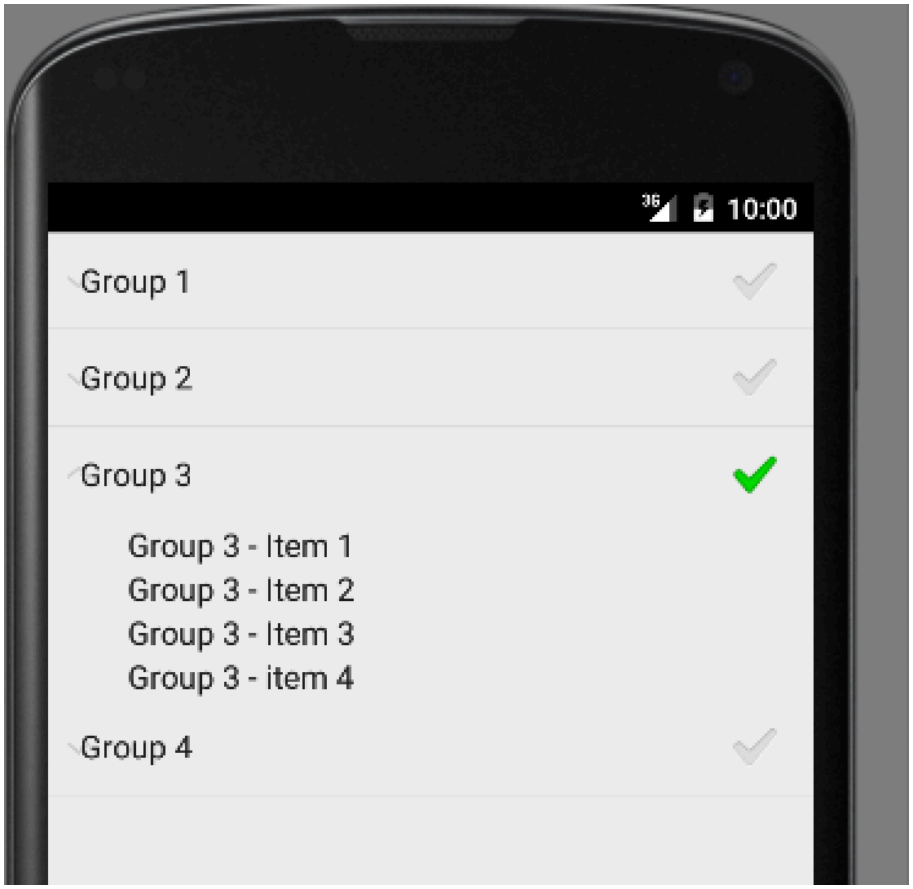
```
@Override
public View getView(int position,
        View convertView, ViewGroup parent) {
  LayoutInflater inflater = LayoutInflater.from(context);
  View row = inflater.inflate(
        android.R.layout.simple_spinner_item, null);
  LanguageInfo info = dataProvider.get(position);
  TextView textView = (TextView)
        row.findViewById(androi
  textView.setText(info.getName(
  return row;
}
```

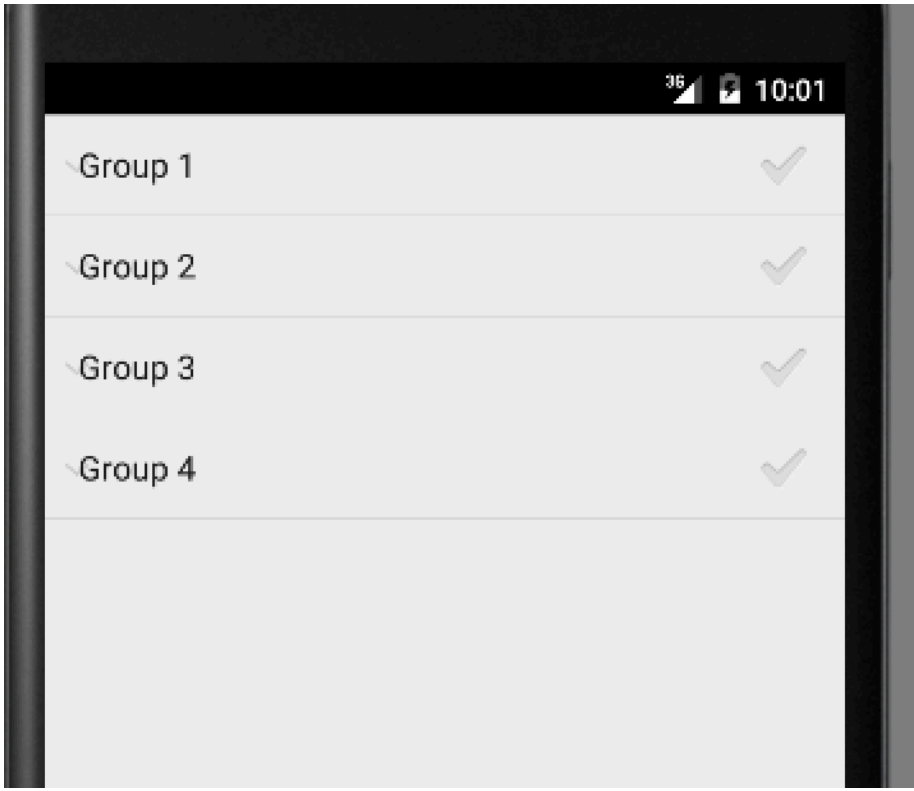


ListView Widget

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  - Simple Spinner
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  - Custom Spinner 2
- **ExpandableListView**

ExpandableListView





ExpandableListView (ExpandableList.java)

```
public class ExpandableList extends ExpandableListActivity {

    private static ArrayList<String> groups;
    private static ArrayList<ArrayList<String>> groupItems;
    private MyExpandableListAdapter adapter;
```

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical" android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ExpandableListView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@android:id/list" />
</LinearLayout>
```

ExpandableListView (ExpandableList.java – static block)

```
static {
    groups = new ArrayList<>(Arrays.asList("Group 1", "Group 2",
            "Group 3", "Group 4"));
    groupItems = new ArrayList<>();
    groupItems.add(new ArrayList<String>(Arrays.asList(
            "Group 1 - Item 1", "Group 1 - Item 2",
            "Group 1 - Item 3", "Group 1 - item 4")));
    groupItems.add(new ArrayList<String>(Arrays.asList(
            "Group 2 - Item 1", "Group 2 - Item 2",
            "Group 2 - Item 3", "Group 2 - item 4")));
    groupItems.add(new ArrayList<String>(Arrays.asList(
            "Group 3 - Item 1", "Group 3 - Item 2",
            "Group 3 - Item 3", "Group 3 - item 4")));
    groupItems.add(new ArrayList<String>(Arrays.asList(
            "Group 4 - Item 1", "Group 4 - Item 2",
            "Group 4 - Item 3", "Group 4 - item 4")));
}
```

ExpandableListView (ExpandableList.java – onCreate)

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_expandable_list);

    adapter = new MyExpandableListAdapter(this, groups, groupItems);
    this.setListAdapter(adapter);
}
```

```
public class MyExpandableListAdapter extends BaseExpandableListAdapter{
    private Context context;
    private ArrayList<String> groups;
    private ArrayList<ArrayList<String>> groupItems;

    public MyExpandableListAdapter(Context context,
                                   ArrayList<String> groups,
                                   ArrayList<ArrayList<String>> groupItems) {
        this.context = context;
        this.groups = groups;
        this.groupItems = groupItems;
    }
```

ExpandableListView (Adapter)

```
@Override
public View getGroupView(int groupPosition,
                         boolean isExpanded,
                         View convertView,
                         ViewGroup parent) {
    LayoutInflater inflater = LayoutInflater.from(context);
    CheckedTextView row = (CheckedTextView)
            inflater.inflate(android.R.layout.simple_list_item_checked, null);
    row.setChecked(isExpanded);
    row.setText(groups.get(groupPosition));
    return row;
}
```

```
@Override
public View getChildView(int groupPosition,
                         int childPosition,
                         boolean isLastChild,
                         View convertView,
                         ViewGroup parent) {
    LayoutInflater inflater = LayoutInflater.from(context);
    TextView row = (TextView)
            inflater.inflate(android.R.layout.simple_expandable_list_item_1, null);
    row.setText(groupItems.get(groupPosition).get(childPosition));
    return row;
}
```